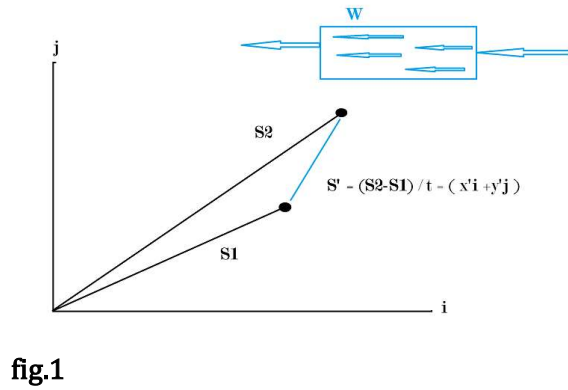


$$K_0 + U_0 = \frac{m}{2} [\dot{x}^2 \mathbf{i} - \dot{y}^2 \mathbf{j}] - \frac{1}{2} C_d A \rho ||\mathbf{r}||^2 ||\mathbf{v}|| t \frac{[\dot{x} \mathbf{i} + \dot{y} \mathbf{j}]}{||\mathbf{v}||} + \rho \left[ \pi r^2 ||\boldsymbol{\varphi}|| t - \frac{4\pi r^3}{6} \right] \frac{||\boldsymbol{\varphi}||^2}{2} \frac{\boldsymbol{\varphi}}{||\boldsymbol{\varphi}||} - \frac{m M_E G}{r + r_E} \mathbf{r}$$



To the left is a graph for the S-frame. S1 and S2 are two observed vectors of an object from the origin S0. It can be seen the translation of the object position is set within the (i, j) coordinate frame. W is a uniform vector field which induces resistance relative to the motion of the object. *This graph accommodates well over small distances where the gravitational field direction changes are small.*

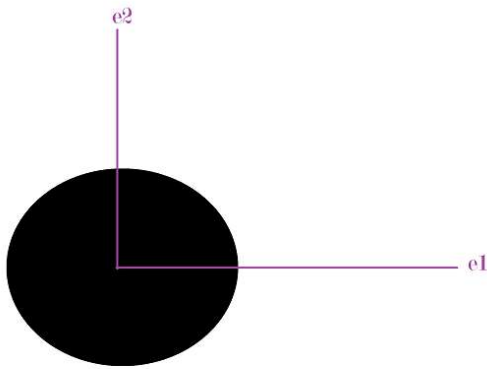


fig.2(on the left) displays the inertial frame of the object.

One way to consider the retrospect for both frames is to say that:

The S-frame (or Static frame) is the observers perspective on the object. Although the origin translates, the basis is parallel to the (i,j) frame.

The I-frame (Inertial frame) is the perspective from the objects point of view with respect to the direction of motion. (i.e. eyes pointed towards 'e1')

The first challenge to this equation is affirming that the S-frame and the I-frame are transposable. This is to insure that the energy equation is consistent for all frames, but evaluated within only one frame, which in this case the equation will be evaluated using the Static Frame.

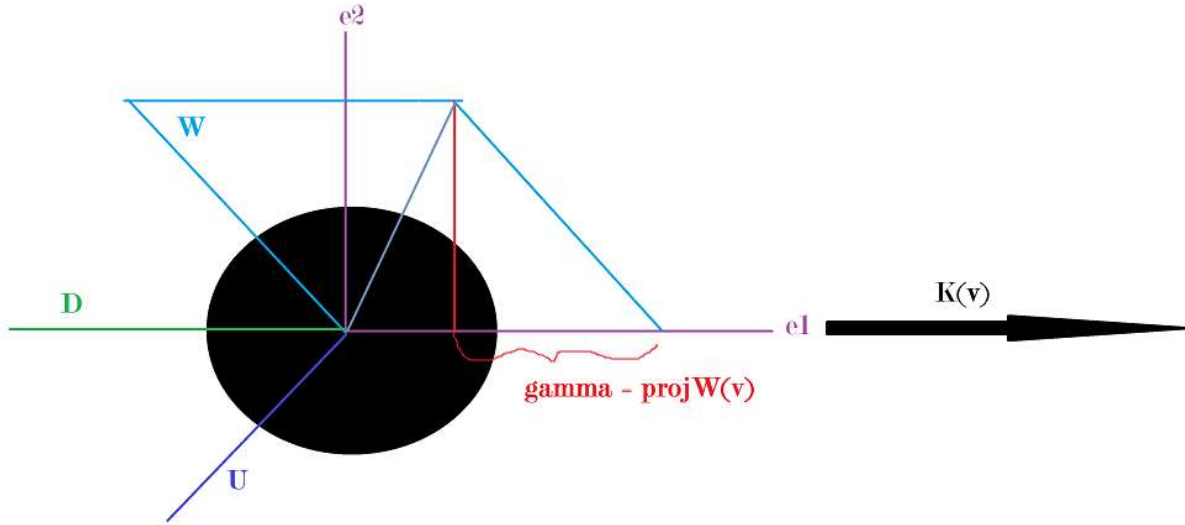


fig.3

The figure above shows the complete inertial free body diagram for the object.

The summary for the vectors is as follows:

$$\mathbf{v} = [\dot{x}\mathbf{i} + \dot{y}\mathbf{j}] = \dot{\mathbf{r}}$$

$\mathbf{K}(\mathbf{v}) = \frac{m}{2} [\dot{x}^2\mathbf{i} + \dot{y}^2\mathbf{j}]$ ; The Kinetic Energy vector dependent on the Mass and Velocity of the object

$\mathbf{U} = \frac{mM_E G}{r+r_E} \mathbf{e}_1, \mathbf{e}_2$ ; The Gravitational energy of the object, although it is always pointed towards the origin of the source of gravity, this vector may assume an arbitrary direction in the Inertial Frame.

$\mathbf{W}$  - The Kinetic Energy vector for the induced resistance, which may assume an arbitrary direction of its own depending on the direction of  $\mathbf{K}(\mathbf{v})$ .

$\mathbf{D}(\text{gamma}) = \gamma = \text{proj}_{\mathbf{v}} \mathbf{W}$ ; the induced drag spanning the  $\mathbf{K}(\mathbf{v})$  component of the object ( usually directly opposite of  $\mathbf{K}(\mathbf{v})$  ).

*Bear in mind that  $\mathbf{W}$  and  $\mathbf{v}$  are essentially velocity vectors. When coupled with mass and the appropriate operations they yield the energy vectors seen in the equation.*

$\mathbf{e}_1 = \frac{\mathbf{v}}{\|\mathbf{v}\|}$ ; the forward motion vector of the object (eyes forward component) in the inertial frame

$\mathbf{e}_2 = \frac{\boldsymbol{\varphi}}{\|\boldsymbol{\varphi}\|}$ ; the orthogonal (ears to ears) component of the object likewise in the inertial frame.

A Review over Each Term in the Equation:

- $\frac{m}{2} [\dot{x}^2 \mathbf{i} + \dot{y}^2 \mathbf{j}]$ ; The instantaneous kinetic energy of the object.
- $\frac{1}{2} C_d A \rho ||\boldsymbol{\gamma}||^2 ||\mathbf{v}|| t \frac{[\dot{x}\mathbf{i} + \dot{y}\mathbf{j}]}{||\mathbf{v}||}$ ; The induced drag energy upon the object along the  $\mathbf{e}_1$  or  $\mathbf{K}(\mathbf{v})$  component.

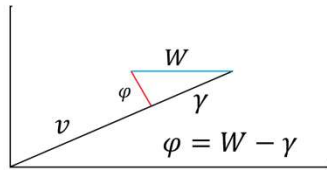
This quantity can be rewritten as  $\frac{1}{2} C_d A \rho ||\boldsymbol{\gamma}||^2 ||\mathbf{v}|| t \frac{[v]}{||\mathbf{v}||} \rightarrow \frac{1}{2} C_d A \rho ||\boldsymbol{\gamma}||^2 ||\mathbf{v}|| t \mathbf{e}_1$

Where  $\frac{1}{2} C_d A \rho ||\boldsymbol{\gamma}||^2$  is the force induced onto the object with respect to the relative velocity of the field  $\mathbf{W}$ . Multiplying by ' $||\mathbf{v}|| t$ ' transforms this into the work done onto the object along the resultant trajectory, and once again multiplying by ' $\mathbf{e}_1$ ' places this vector along the span of  $\mathbf{K}(\mathbf{v})$  and into the static frame. Thus the Force/Work portion is the scalar that is then transposed into a vector quantity through the use of ' $\mathbf{e}_1$ '.

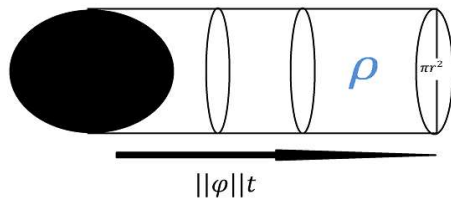
++ The argument for the use of ' $\mathbf{v}$ ' instead of ' $\boldsymbol{\delta} = \mathbf{W} + \mathbf{v}$ ', is that ' $\mathbf{v}$ ' should be the resultant velocity of the object which lends a hand to solving the equations of motion. While ' $\boldsymbol{\gamma}$ ' traces the force along the span of  $\mathbf{K}(\mathbf{v})$ . The quantity accommodates the ' $\mathbf{e}_1$ ' contribution of energy while this next term (with volume) accounts for the ' $\mathbf{e}_2$ ' contribution.

- $\rho \left[ \pi r^2 ||\boldsymbol{\phi}|| t - \frac{4\pi r^3}{6} \right] \frac{||\boldsymbol{\phi}||^2}{2} \frac{\boldsymbol{\phi}}{||\boldsymbol{\phi}||}$ ; This quantity considers the surface area of the object as well as the mass flow rate imparted by the field  $\mathbf{W}$ .

$\boldsymbol{\phi} = \mathbf{W} - \boldsymbol{\gamma}$ ; Is the  $\mathbf{e}_2$  component imparted by  $\mathbf{W}$ .

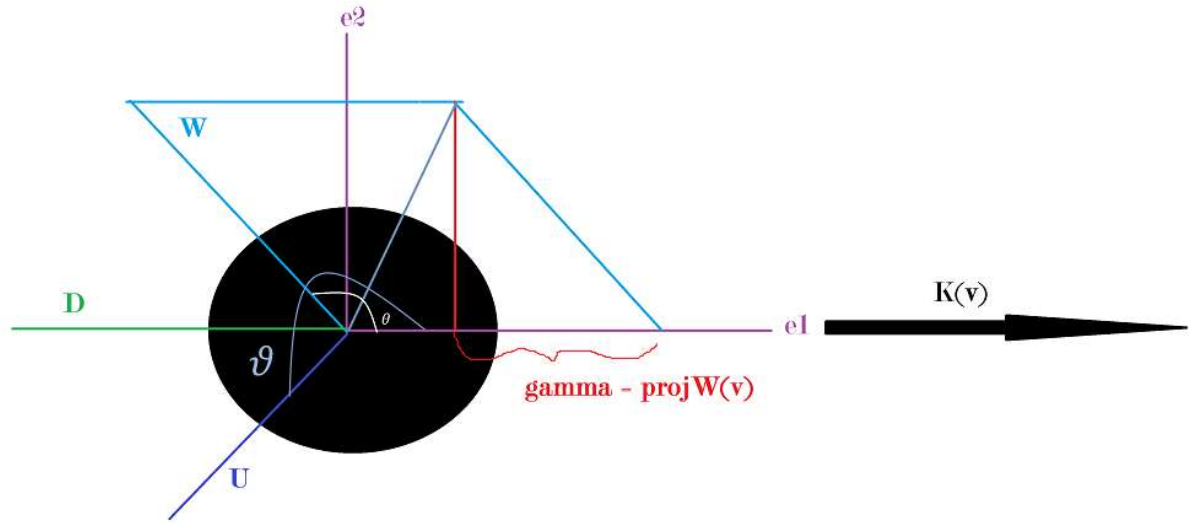


In the picture below, the object continuously covers a certain volume per time. Conform this mass flow to the shape of the object, a sphere in this case; to yield the volume of the cylinder encompassed by the objects motion minus half the spheres volume since the air impacting the object only affects half its surface.



Thus far the vectors in the inertial frame have been translated into the static (observers) frame.

# Force Translation via The Lagrangian



$$E = K_0 + U_0 = \frac{m}{2} ||v||^2 \mathbf{e}_1 - \frac{1}{2} C_d A ||\gamma||^2 ||v|| t \mathbf{e}_1 + \rho \left[ \pi r^2 ||\phi|| t - \frac{4\pi r^3}{6} \right] \frac{||\phi||^2}{2} \mathbf{e}_2 - \frac{m M_E G}{r + r_E} \cos(\vartheta) \mathbf{e}_1 - \frac{m M_E G}{r + r_E} \sin(\vartheta) \mathbf{e}_2$$

As you can see the energy equation has been rewritten under the object basis.  $\mathbf{W} = W \cos \theta \mathbf{e}_1 + W \sin \theta \mathbf{e}_2$ ;  $\gamma = W \cos \theta \mathbf{e}_1$ ,  $\phi = W \sin \theta \mathbf{e}_2$

$$L = E ;$$

$$\frac{\partial L}{\partial \mathbf{r}} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{r}}} = 0$$

$$\frac{\partial L}{\partial r} = -\frac{m M_E G}{(r + r_E)^2} \cos(\vartheta) \mathbf{e}_1 - \frac{m M_E G}{(r + r_E)^2} \sin(\vartheta) \mathbf{e}_2$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{r}}} = m \dot{v} \mathbf{e}_1 - \frac{1}{2} C_d A \rho ||\gamma||^2 \mathbf{e}_1$$

This shows that the forces are acting under the correct assumptions:

\*\*That the **drag force** is derived correctly and **acts along the span of the velocity vector**

\*\*The **gravitational force** is derived correctly, and it **acts with respect to the basis vectors**.

Therefore the Lagrangian operator proves that the energy equation is sensible. Since the sum of the Forces includes only the gravitational and total drag forces.

Computing the magnitudes for each term in the Lagrangian shows this more clearly.

$$\left| \frac{\partial L}{\partial \mathbf{r}} \right| = - \frac{m M_E G}{(r + r_E)^2} \mathbf{r}_e$$

$$\left| \frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{r}}} \right| = \left( m \dot{v} - \frac{1}{2} C_d A \rho |\mathbf{v}|^2 \right) \mathbf{r}_s$$

$\left| \frac{\partial L}{\partial \mathbf{r}} \right| - \left| \frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{r}}} \right| = 0$  is true when  $\mathbf{r}_s, \mathbf{r}_e$  share the same span, where  $\mathbf{r}_s, \mathbf{r}_e$  are the basis for this Lagrangian operator. This shows that the system is non-conservative. It may be argued that at each instance  $\mathbf{r}_s, \mathbf{r}_e$  share the same span.

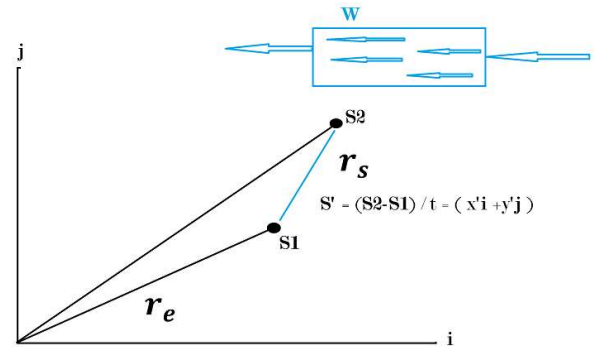


fig.1a

**Setting the acceleration to zero**, shows that the gravitational force can be equal to the drag force induced by the field  $\mathbf{W}$ , since the gravity field is always along the radial vector, there can be only certain locations in the  $(\mathbf{i}, \mathbf{j})$  frame where this is possible (depending on the speed of  $\mathbf{W}$ ). In **fig.1a** the only possible location is along the span of  $\langle -\mathbf{i}, 0 \rangle$ .

*In a  $\mathbf{W}$  field that is static*, the relative velocity of the drag energy is always zero, and the magnitude of the drag force is directly proportional with the density of the fluid and motion/mass of the object. In affect all space is available for the static case. Yielding...

$$- \frac{m M_E G}{(r + r_E)^2} = m \dot{v} - \frac{1}{2} C_d A \rho v^2$$

In the span of the  $\mathbf{S}$  frame, since  $\mathbf{e}_1 = \frac{\mathbf{v}}{||\mathbf{v}||} = \frac{\Delta \mathbf{S}}{t ||\mathbf{v}||}$

$$\Delta ||\mathbf{S}|| = \Delta r$$

$$\Delta ||\mathbf{v}|| = \dot{r}$$

Front point to point in the  $(\mathbf{i}, \mathbf{j})$  frame, each change in position is a change in 'r' within the  $\mathbf{S}$  frame. At each instance, the object returns to the origin in the  $\mathbf{S}$  frame and changes 'r' towards the next instance.

**This lends a hand to the appropriateness for the Lagrangian.**

From now on, the  $\mathbf{W}$  field will be static, for simplicity.

$m \mathbf{v}'$  is the resultant path the object takes. Which means the next task is to decipher the thrust vector.